

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

11

10 answers · Rationale-rich · Teach from doc

Q1**A****A.** -152

Choice A is correct. The value of $f(0)$ is the value of $f(x)$ when $x = 0$. The function f is defined by $f(x) = 10x^2 - 32x - 152$. Substituting 0 for x in this equation yields $f(0) = 10(0)^2 - 32(0) - 152$. This equation can be rewritten as $f(0) = 10(0) - 0 - 152$, or $f(0) = -152$. Therefore, the value of $f(0)$ is -152.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**A****A.** 3

Choice A is correct. Adding 40 to both sides of the given equation yields $8x^2 = 72$. Dividing both sides of this equation by 8 yields $x^2 = 9$. Taking the square root of both sides of this equation yields $x = 3$ or $x = -3$. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect. This is the solution to the equation $8x = 32$, not the given equation.

Choice C is incorrect. This is the solution to the equation $8x - 40 = 32$, not the given equation.

Choice D is incorrect. This is the solution to the equation $x - 40 = 32$, not the given equation.

Q3**D**

D. $2a^3 + 6a^2$

Choice D is correct. Expanding the given expression using the distributive property yields $2a^2(a) + 2a^2(3)$. Combining like terms yields $2a^2(a^1) + (2 \times 3)(a^2)$, or $2a^{2+1} + 6a^2$, which is equivalent to $2a^3 + 6a^2$.

Choices A and B are incorrect and may result from incorrectly combining like terms. Choice C is incorrect and may result from distributing $2a^2$ only to a , and not to 3, in the given expression.

Q4**6**

The correct answer is 6. An x -intercept of a graph is a point on the graph where it intersects the x -axis, or where the value of y is 0. The graph shown intersects the x -axis at the point 6, 0. Therefore, the x -coordinate of the x -intercept of the graph shown is 6.

Q5**D**

D. -3

Choice D is correct. If $(x+5)^2 = 4$, then taking the square root of each side of the equation gives $x+5 = 2$ or $x+5 = -2$. Solving these equations for x gives $x = -3$ or $x = -7$. Of these, -3 is the only solution given as a choice.

Choice A is incorrect and may result from solving the equation $x+5 = 4$ and making a sign error. Choice B is incorrect and may result from solving the equation $x+5 = 4$. Choice C is incorrect and may result from finding a possible value of $x+5$.

Q6**D**D. $2(x^3 + 2)$

Choice D is correct. The expression $2x^3 + 4$ has two terms, $2x^3$ and 4. The greatest common factor of these two terms is 2. Factoring 2 from each of these terms yields $2(x^3) + 2(2)$, or $2(x^3 + 2)$.

Choices A and B are incorrect because 4 is not a factor of the term $2x^3$. Choice C is incorrect and may result from factoring 2 from $2x^3$ but not from 4.

Q7**B**

B. 16

Choice B is correct. The value of $f144$ is the value of f_x when $x = 144$. It's given that the function f is defined by $f_x = 4 + \sqrt{x}$. Substituting 144 for x in this equation yields $f144 = 4 + \sqrt{144}$. Since the positive square root of 144 is 12, it follows that this equation can be rewritten as $f144 = 4 + 12$, or $f144 = 16$. Therefore, the value of $f144$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f1,296$, not $f144$.

Choice D is incorrect. This is the value of $f5,184$, not $f144$.

Q8**A**

A. $c = \frac{b}{42f}$

Choice A is correct. It's given that the equation $b = 42cf$ relates the positive numbers b , c , and f . Dividing each side of the given equation by $42f$ yields $\frac{b}{42f} = c$, or $c = \frac{b}{42f}$. Thus, the equation $c = \frac{b}{42f}$ correctly expresses c in terms of b and f .

Choice B is incorrect. This equation can be rewritten as $b = cf + 42$.

Choice C is incorrect. This equation can be rewritten as $b = \frac{c}{42f}$.

Choice D is incorrect. This equation can be rewritten as $b = 42 - c - f$.

Q9**B**

B. $3x + 2$

Choice B is correct. Using the associative and commutative properties of addition, the given expression $(x + 5) + (2x - 3)$ can be rewritten as $(x + 2x) + (5 - 3)$. Adding these like terms results in $3x + 2$.

Choice A is incorrect and may result from adding $(x - 5) + (2x + 3)$. Choice C is incorrect and may result from adding $(x - 5) + (2x - 3)$. Choice D is incorrect and may result from adding $(x + 5) + (2x + 3)$.

Q10

B

B.

x	1	2	3
hx	-2	1	6

Choice B is correct. It's given that $hx = x^2 - 3$. Each table gives 1, 2, and 3 as the three given values of x . Substituting 1 for x in the equation $hx = x^2 - 3$ yields $h1 = 1^2 - 3$, or $h1 = -2$. Substituting 2 for x in the equation $hx = x^2 - 3$ yields $h2 = 2^2 - 3$, or $h2 = 1$. Finally, substituting 3 for x in the equation $hx = x^2 - 3$ yields $h3 = 3^2 - 3$, or $h3 = 6$. Therefore, hx is -2 when x is 1, hx is 1 when x is 2, and hx is 6 when x is 3. Choice B is a table with these values of x and their corresponding values of hx .

Choice A is incorrect. This is a table of values for the function $hx = x + 3$, not $hx = x^2 - 3$.

Choice C is incorrect. This is a table of values for the function $hx = 2x - 3$, not $hx = x^2 - 3$.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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